



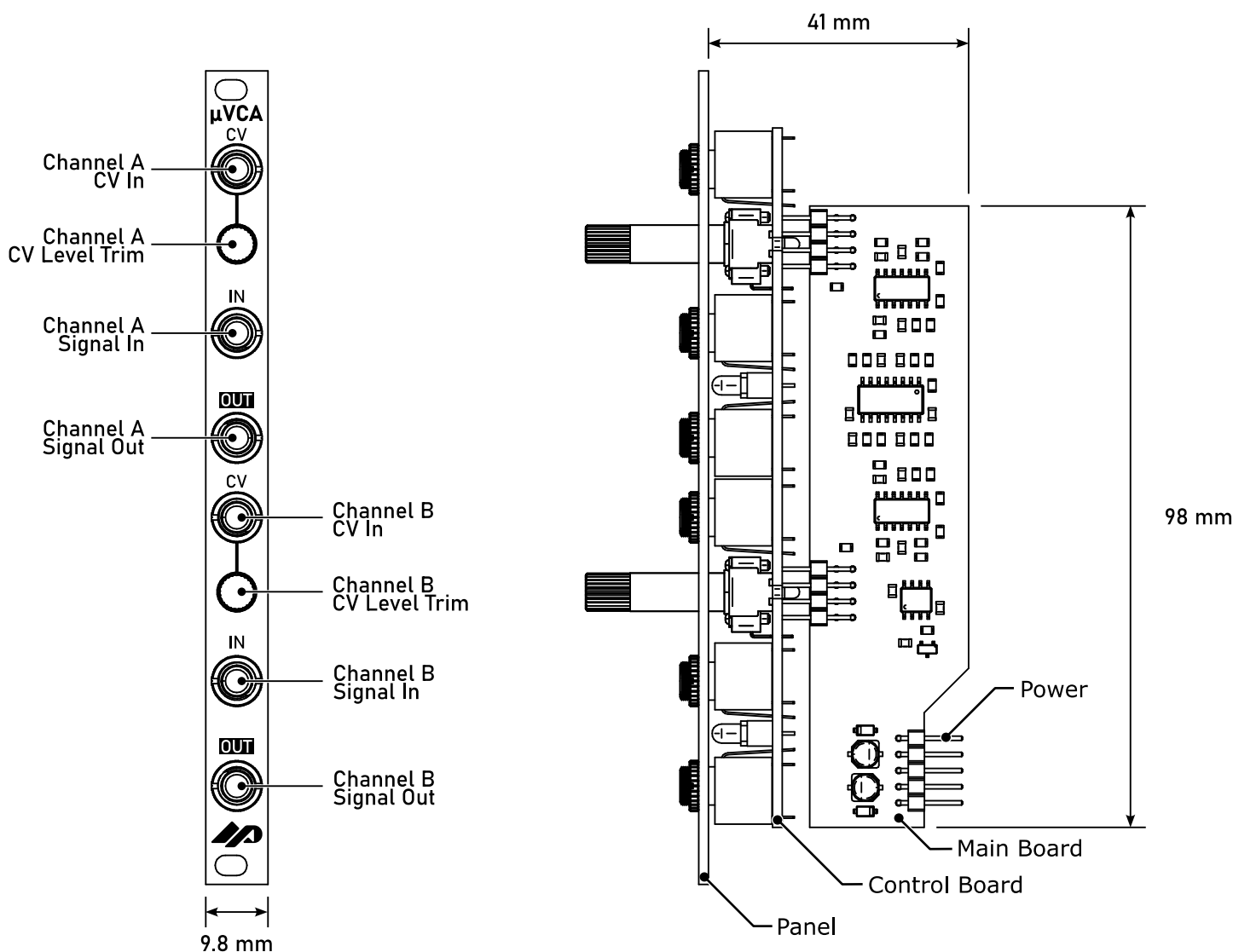
Description

The uVCA is a dual-channel SSI2164-based linear VCA compressed into 2hp. Input signals can be attenuated down to -100dB using a standard 0-5V CV input voltage range or amplified up to +6dB with a 10V CV input. Each VCA channel signal input is DC-coupled and can process both audio and CV signals.

Features

- SSI2164-based architecture
- 2 independent DC-coupled VCA channels
- Linearized CV response
- CV level trim potentiometers
- Bipolar indicator LEDs for each signal output
- Current Draw: +12V: 42mA, -12V: 33mA

Module Layout



Functionality

Signal Inputs/Outputs:

The signal inputs of Channel A and Channel B are DC coupled and can accept both CV and audio signals. Input signals can range from -10V and +10V. Larger signals may cause unwanted effects or distortion but will not damage the module.

Signal output level is indicated by the bicolor LED above the jack. A green color indicates positive output voltage, while red indicates negative output voltage.

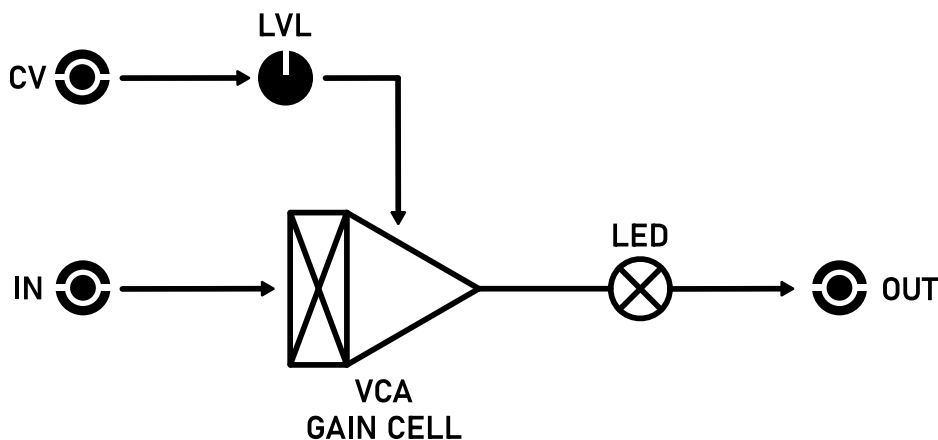
CV Inputs:

CV input levels can range from 0-10V: 0V will provide a gain of -100dB, 5V will provide unity (+0dB) gain, and 10V will provide +6dB gain. A negative CV signal has no further effect on reducing signal gain.

CV Level Trim:

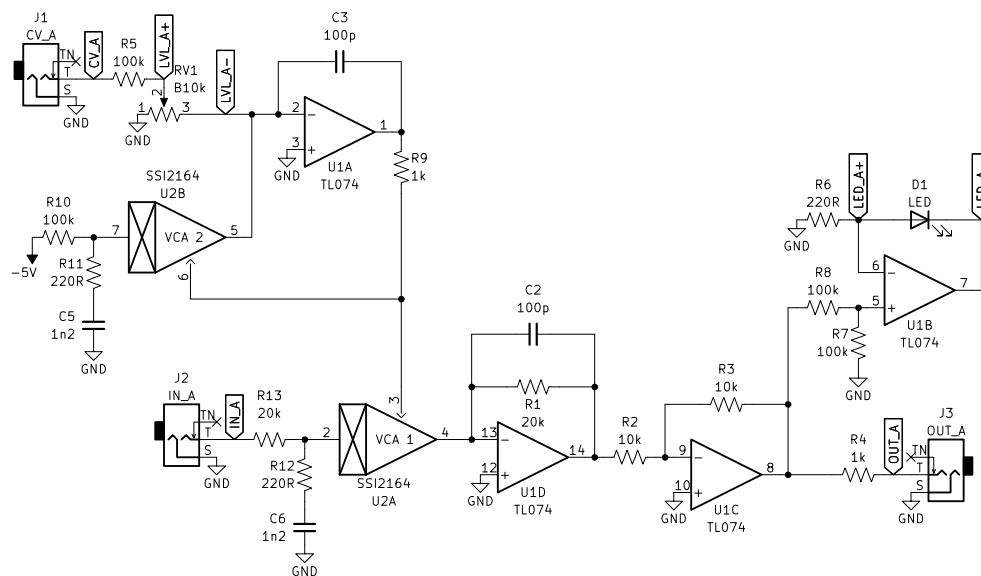
Each CV input can be independently trimmed using the Level trim knobs. Both CV trim knobs act as simple attenuators.

Block Diagram (Single Channel)



Bill of Materials

Value	Qty.	Name on Board	Description	Footprint
220Ω	6	R6, R11, R12, R15, R17, R22	220Ω 1% 0.125W Resistor	0805_R
1kΩ	4	R4, R9, R19, R23	1kΩ 1% 0.125W Resistor	0805_R
10kΩ	4	R2, R3, R25, R27	10kΩ 1% 0.125W Resistor	0805_R
20kΩ	4	R1, R13, R18, R24	20kΩ 1% 0.125W Resistor	0805_R
27kΩ	2	R14, R28	27kΩ 1% 0.125W Resistor	0805_R
100kΩ	8	R5, R7, R8, R10, R16, R20, R21, R26	100kΩ 1% 0.125W Resistor	0805_R
100pF	4	C2, C3, C12, C14	100pF 50V C0G MLCC	0805_C
100nF	9	C1, C4, C7, C8, C11, C13, C15, C16, C17	100nF 50V X7R MLCC	0805_C
10uF	2	C18, C19	10uF 25V Elec. Capacitor	SMD,4x5.7
Schottky Diode	2	D3, D4	Schottky Diode 40V	SOD-123
TL074	2	U1, U3	TL074 Quad Op Amp	SOIC-14,3.9x8.7mm_P1.27mm
TL072	1	U4	TL072 Dual Op Amp	SOIC-8,3.9x4.9mm_P1.27mm
LM4040	1	U5	LM4040 5V Reference	SOT-23
B10k Pot Lin	2	RV1, RV2	10k Lin. Trim Pot	RK09K
3.5mm Jack	6	J1, J2, J3, J4, J5, J6	3.5mm Mono Jack	PJ-3001F
3mm LED	2	D1, D2	3mm LED Bipolar Red/Green	LED 3mm
4mm LED Spacer	2	N/A	3mm x 4mm H LED Spacer	N/A
8-Pin Header	2	J7/J9, J8/J10	2.54mm Header 4x2	2.54mm 4x2 Right Angle
5-Pin Header	1	J11	2.54mm Header 5x1	2.54mm 5x1 Right Angle



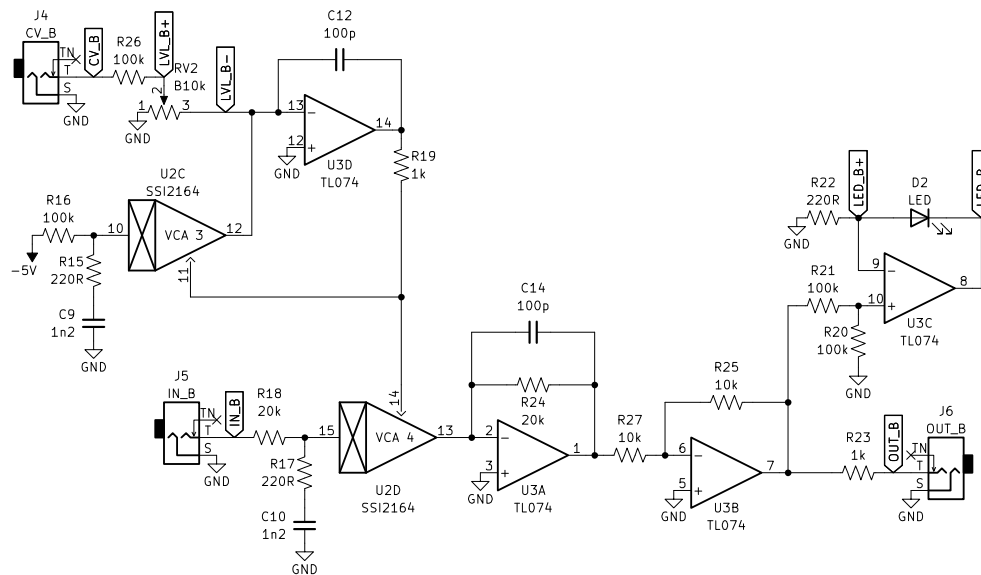
Drachma Mfg.

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Size: A4 Date:
KiCad E.D.A. 10.0.3

Rev: 1
Id: 1/5



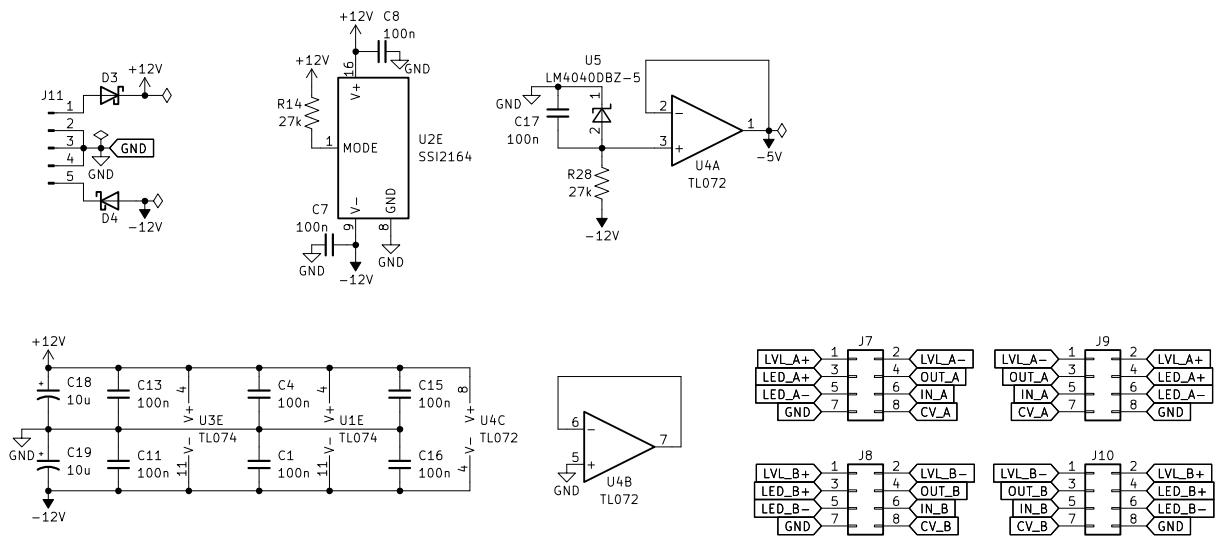
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